

Dynamic Factor Investing with Balanced Temporal Factor Autoencoder

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Abstract

Static factor models such as Fama-French and HXZ assume linearity and temporal stationarity, assumptions that are frequently violated in dynamic markets like China’s A-shares. While Transformer-based models demonstrate superior predictive accuracy in capturing time-varying factor premia, they suffer from large drawdowns and limited interpretability, hindering practical deployment. We introduce the Temporal Factor Autoencoder (TFA), which extends the Transformer architecture through multi-task learning with economically-motivated regularization. TFA incorporates three key components: reconstruction loss to ground latent factors in observable characteristics, smoothness regularization for temporal stability, and orthogonality constraints to ensure factor independence. Through systematic ablation studies and hyperparameter tuning, we identify an optimal configuration that achieves 66% lower maximum drawdown (0.1239 vs 0.3613) compared to the standard Transformer while maintaining competitive Sharpe ratio (0.8759 vs 0.9348, only 6% decrease). Our results demonstrate the practical value of incorporating economic structure into machine learning models for factor investing, offering institutional investors a compelling risk-return profile suitable for drawdown-constrained strategies.

1 Introduction

1.1 Motivation: The Limits of Static Factor Models

Canonical factor models, including the Fama-French three-factor [1] and five-factor models [2], as well as the Hou-Xue-Zhang q-factor model [4], have been foundational in asset pricing research. These models typically assume linearity in the relationship between firm characteristics and expected returns, as well as temporal stationarity in factor loadings and premia. However, these assumptions are frequently violated in dynamic markets, particularly in emerging economies like China’s A-share market.

The Chinese stock market is characterized by frequent regime shifts driven by regulatory changes, evolving market microstructure, and rapid economic development. Factor premia exhibit significant time variation, with periods of dominance alternating unpredictably across valuation, momentum, and quality factors. Static models fail to capture these dynamic relationships, leading to diminished out-of-sample predictive power and suboptimal portfolio construction. This limitation motivates the need for adaptive modeling frameworks that can accommodate time-varying factor structures while maintaining economic interpretability.

1.2 Research Objective

This paper aims to improve cross-sectional return prediction in the CSI 500 universe through a hybrid machine learning pipeline that combines classical dimensionality reduction with modern sequence modeling. Our approach encompasses four key components: (1) dynamic factor extraction via rolling Principal Component Analysis (PCA) to create adaptive, denoised features; (2) sequence modeling using Transformer encoders to capture temporal dependencies; (3) an interpretable Temporal Factor Autoencoder (TFA) architecture as our core contribution; and (4) rigorous out-of-sample evaluation using a rolling-window protocol with realistic transaction costs.

While Transformer models demonstrate strong predictive performance, they present two critical limitations for practical deployment: large maximum drawdowns indicating portfolio instability, and a "black box" nature that provides limited insight into the economic drivers of predictions. Our central research question is: *Can we maintain Transformer’s predictive power while significantly improving stability and interpretability through economically-motivated architectural constraints?*

1.3 Key Empirical Findings

Our empirical investigation reveals that Transformer models achieve superior predictive accuracy (Information Coefficient of 0.0361, Sharpe ratio of 0.9348) compared to linear and standard deep learning baselines. However, they suffer from a maximum drawdown of 36.13%, which poses significant practical constraints for institutional deployment. Through systematic experimentation, we demonstrate that TFA with an optimally tuned reconstruction weight parameter ($\alpha = 0.02$) achieves 66% lower maximum drawdown (0.1239 vs 0.3613) while sacrificing only 6% of the Sharpe ratio (0.8759 vs 0.9348). Ablation studies validate that each component of the multi-task loss function—reconstruction, smoothness, and orthogonality—serves a distinct economic purpose and contributes to model stability.

1.4 Paper Structure

The remainder of this paper is organized as follows. Section 2 reviews relevant literature on factor models, machine learning in asset pricing, and autoencoder architectures. Section 3 describes our methodology, including data preprocessing, feature engineering via rolling PCA, baseline model specifications, and the TFA architecture. Section 4 presents our empirical results, progressing from baseline comparisons through ablation studies to our final optimized configuration. Section 5 discusses implications for the Chinese stock market and broader methodological contributions. Section 6 concludes with directions for future research.

2 Literature Review

Our work builds on three streams of literature: traditional factor models, machine learning applications in asset pricing, and autoencoder-based approaches for financial data.

Factor Models. The factor model paradigm, pioneered by [1] and extended by [2], has been central to empirical asset pricing. [4] propose the q-factor model emphasizing investment and profitability factors. These models assume static, linear relationships between firm characteristics and returns. While parsimonious and interpretable, they struggle to accommodate the nonlinear dynamics and regime-dependent behavior observed in emerging markets.

Machine Learning in Asset Pricing. Gu et al. [3] demonstrate that neural networks significantly outperform linear models in predicting individual stock returns, achieving substantial improvements in out-of-sample R^2 . Their work establishes the potential for machine learning to capture complex, nonlinear relationships in asset returns. However, their models lack temporal structure and economic interpretability, limiting practical adoption by insti-

tutional investors who require explainable predictions for risk management and regulatory compliance.

Autoencoders in Finance. Kelly et al. [5] introduce instrumented PCA, using autoencoders to extract latent factors from firm characteristics. Their approach demonstrates that constraining latent representations to reconstruct input features can improve economic interpretability. However, their framework is static and does not account for temporal evolution in factor structures. Vaswani et al. [6] introduce the Transformer architecture with self-attention mechanisms, which has proven effective for sequence modeling across domains but remains largely a "black box" without economic constraints.

Research Gap. Existing machine learning models for asset pricing either lack temporal structure (e.g., feed-forward networks) or lack economic interpretability (e.g., standard Transformers). Our contribution is to bridge this gap by introducing TFA, which combines Transformer-based sequence modeling with economically-motivated regularization through multi-task learning. This allows us to maintain predictive performance while significantly improving portfolio stability and providing interpretable factor structures.

3 Methodology

3.1 Data Universe and Preprocessing

Universe. Our analysis focuses on the CSI 500 Index constituents, representing medium-capitalization firms in China’s A-share market. This universe provides sufficient cross-sectional diversity while maintaining reasonable liquidity for portfolio implementation.

Sample Period. We use monthly data from January 2013 to December 2023, providing approximately 132 months of observations. This period captures multiple market regimes, including the 2015 bubble and crash, the 2018 trade war impact, and the post-pandemic recovery.

Firm Characteristics. We construct a comprehensive set of firm-level characteristics spanning four categories: (1) Valuation metrics including price-to-earnings (P/E) and price-to-book (P/B) ratios; (2) Profitability and quality measures including return on equity and operating profit margin; (3) Leverage and liquidity indicators; and (4) Growth and sentiment variables including sales growth and analyst recommendations.

Preprocessing Pipeline. Accounting data are forward-filled to reflect information availability, with monthly alignment ensuring temporal consistency. We filter out stocks with less than 24 months of historical data to ensure stable estimation. For each cross-section, we apply three transformations: (1) Winsorization at the 1st and 99th percentiles

to mitigate extreme outliers; (2) Cross-sectional z-score normalization to standardize feature scales; (3) Neutralization of size and industry effects through orthogonal regression, isolating characteristic-specific variation.

3.2 Feature Engineering: Dynamic Factors via Rolling PCA

Traditional PCA assumes a static covariance structure, which is inappropriate for time-varying markets. We implement rolling-window PCA to create adaptive factor representations that evolve with market conditions.

Mechanism. For each rebalancing date t , we estimate PCA on a 24-month trailing window of processed characteristics. We drop zero-variance features within each window and apply two constraints: a hard component cap of $N_{\text{components}} = 10$ and a soft variance threshold of 80%. This procedure typically extracts 11 principal components explaining approximately 42% of cross-sectional variance.

Rationale. Rolling PCA serves three purposes: (1) Dimensionality reduction from dozens of potentially collinear characteristics to a parsimonious set of orthogonal factors; (2) Noise reduction by filtering out idiosyncratic variation while preserving systematic patterns; (3) Adaptivity through time-varying factor loadings that respond to evolving market dynamics. The moderate explained variance (42%) reflects our conservative approach, prioritizing robust factors over exhaustive variance capture.

Factor Score Construction. After fitting PCA on the trailing window, we project month t characteristics onto the learned components to obtain factor scores for each stock. We stack the most recent 12 months of factor scores to form sequences, forward-filling missing values up to 3 months. This produces a 3D input tensor of shape $(N_{\text{stocks}}, T = 12, K_{\text{PCA}} = 11)$ for model training.

3.3 Baseline Model Architectures

To establish performance benchmarks, we implement four baseline models spanning linear, ensemble, and deep learning approaches.

Ridge Regression. Our linear baseline applies L2 regularization to prevent overfitting in high-dimensional settings. Ridge regression provides interpretable coefficients and serves as a robust benchmark for more complex nonlinear models.

Random Forest. This ensemble method captures nonlinear relationships through recursive partitioning without requiring explicit temporal structure. We use 100 trees with maximum depth of 10 to balance expressive power and generalization.

Multi-Layer Perceptron (MLP). As a standard deep learning baseline, MLP with three hidden layers (256, 128, 64 units) tests whether complexity alone, without temporal structure, improves predictions.

Transformer Encoder. Our primary sequence baseline uses multi-head self-attention to model temporal dependencies in factor score sequences. The architecture includes positional encoding, 2 encoder layers with 4 attention heads, and a feedforward dimension of 256. This serves as the key comparison point for evaluating TFA’s improvements.

Evaluation Protocol. All models are trained using a rolling-window scheme: 60 months for training, 12 months for validation, and 1-month-ahead prediction. Hyperparameters are selected via nested walk-forward validation within each training window. We evaluate both predictive metrics (Information Coefficient, IC Information Ratio) and portfolio performance (long-short Sharpe ratio, maximum drawdown, turnover) after 10 basis points transaction costs per side.

3.4 Temporal Factor Autoencoder (TFA)

Motivation. While Transformer models demonstrate strong predictive performance, they present two critical limitations: (1) large portfolio drawdowns indicating insufficient risk control, and (2) opaque ”black box” representations lacking economic interpretability. TFA addresses these issues by extending Transformer with explicit constraints on the learned latent factors.

Architecture. TFA consists of three components operating on factor score sequences $\mathbf{X} \in \mathbb{R}^{N \times T \times K}$:

1. *Encoder*: A Transformer maps input sequences to latent representations $\mathbf{Z} \in \mathbb{R}^{N \times D}$ through self-attention, capturing temporal dependencies.
2. *Decoder*: A separate network reconstructs the original factor scores $\hat{\mathbf{X}}$ from latent factors $\mathbf{Z}$, ensuring information preservation.
3. *Prediction Head*: A linear layer maps latent factors to return forecasts $\hat{r}$.

Multi-Task Loss Function. TFA jointly optimizes four objectives:

$$\mathcal{L}_{\text{TFA}} = \mathcal{L}_{\text{pred}} + \alpha \cdot \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{smooth}} + \gamma \cdot \mathcal{L}_{\text{ortho}} \quad (1)$$

where:

- $\mathcal{L}_{\text{pred}} = \|\hat{r} - r\|^2$ measures prediction accuracy.

- $\mathcal{L}_{\text{recon}} = \|\hat{\mathbf{X}} - \mathbf{X}\|^2$ ensures latent factors preserve information content, grounding them in observable characteristics.
- $\mathcal{L}_{\text{smooth}} = \|\mathbf{Z}_t - \mathbf{Z}_{t-1}\|^2$ penalizes temporal variation in factor loadings, promoting stable factor evolution.
- $\mathcal{L}_{\text{ortho}} = \|\mathbf{Z}^\top \mathbf{Z} - \mathbf{I}\|^2$ encourages factor independence, reducing redundancy.

Design Rationale. Each loss component serves a distinct economic purpose. Reconstruction prevents arbitrary latent representations by requiring factors to encode sufficient information to recover input characteristics, enhancing interpretability. Smoothness regularization reduces erratic period-to-period changes in factor exposures, improving portfolio stability and transaction cost efficiency. Orthogonality ensures factors capture distinct sources of return variation, facilitating risk attribution and improving diversification.

Key Design Insight. The parameter α controls the prediction-risk trade-off: higher values prioritize interpretability and stability at the cost of predictive accuracy, while lower values emphasize prediction. Parameters β and γ provide essential regularization; ablation studies (Section 4.3) confirm that removing them severely degrades model performance. We use hyperparameters $d_{\text{model}} = 64$, $n_{\text{heads}} = 4$, $n_{\text{layers}} = 2$, trained for 50 epochs with learning rate 5×10^{-4} .

4 Empirical Results

4.1 Baseline Comparison: Linear vs Deep Learning

Table 1 presents out-of-sample performance for our four baseline models over the full evaluation period.

Table 1: Baseline Model Performance Comparison

Model	IC	ICIR	Sharpe	MaxDD
Ridge	0.0310	0.2534	0.9459	0.3045
MLP	-0.0056	-0.0614	0.0674	0.2509
Transformer	0.0361	0.2706	0.9348	0.3613

Key Finding. Transformer achieves substantial improvements over both linear and standard deep learning baselines. Compared to Ridge regression, Transformer increases the Information Coefficient by 16% (0.0361 vs 0.0310) and ICIR by 7% (0.2706 vs 0.2534), while achieving comparable Sharpe ratio (0.9348 vs 0.9459) with moderately higher drawdown.

The MLP baseline performs poorly with negative IC (-0.0056), likely due to overfitting without proper temporal structure.

Analysis. Ridge regression provides a solid linear baseline with strong risk-adjusted performance, demonstrating that simple models can be effective when features are well-engineered through PCA. MLP’s poor performance demonstrates that architectural complexity alone, without appropriate inductive biases for sequential data, is insufficient. Transformer’s success validates the importance of temporal modeling through attention mechanisms, which effectively capture dynamic relationships in factor score sequences.

Remaining Challenge. Despite Transformer’s strong predictive performance, its maximum drawdown of 36.13% poses practical concerns for risk-constrained institutional investors. Additionally, the Transformer functions as a “black box,” providing limited insight into which factors drive predictions or how factor importance evolves over time. This raises a critical question: Can we maintain Transformer’s predictive power while dramatically improving stability and interpretability?

4.2 Introducing TFA: Initial Results

To address Transformer’s limitations, we introduce TFA with multi-task learning as described in Section 3.4. Table 2 compares Transformer with TFA using default hyperparameters.

Table 2: Transformer vs TFA (Default Configuration)

Model	IC	ICIR	Sharpe	MaxDD
Transformer	0.0361	0.2706	0.9348	0.3613
TFA (default)	0.0218	0.2450	0.8434	0.2588

Initial Assessment. TFA with default hyperparameters demonstrates a favorable trade-off. Maximum drawdown improves by 28% (0.2588 vs 0.3613), a substantial gain in portfolio stability. However, this comes at a cost: IC decreases by 40% (0.0218 vs 0.0361) and Sharpe ratio declines by 10% (0.8434 vs 0.9348). While these results are promising, they raise a natural question: Is this the optimal configuration, or can we find a better balance through systematic hyperparameter tuning?

4.3 Ablation Study: Understanding TFA Components

Before optimizing hyperparameters, we conduct ablation studies to validate each loss component’s contribution. Table 3 presents results when removing key components.

Table 3: Ablation Study: Testing Loss Components

Configuration	IC	ICIR	Sharpe	MaxDD	Interpretation
Remove Recon. ($\alpha = 0$)	0.0352	0.3085	0.4023	0.2412	IC improves but Sharpe suffers
Remove Smooth/Ortho. ($\beta = \gamma = 0$)	0.0094	0.0697	0.5242	0.2908	Severe prediction degradation

Key Insights. Removing reconstruction ($\alpha = 0$) dramatically improves predictive accuracy (IC increases to 0.0352, ICIR to 0.3085), validating that reconstruction imposes a constraint on the model’s optimization. However, this comes at a significant cost: Sharpe ratio drops to 0.4023, indicating that pure prediction optimization without reconstruction leads to less stable portfolios. Conversely, removing smoothness and orthogonality regularization ($\beta = \gamma = 0$) severely degrades prediction performance (IC drops to 0.0094), while maintaining lower drawdown. This confirms that these components are essential for predictive power and should not be removed.

Conclusion. All loss components serve important purposes. However, the ablation results suggest that α (reconstruction weight) is the primary hyperparameter governing the prediction-risk trade-off. This motivates a systematic search for the optimal α value that maximizes risk-adjusted performance.

4.4 Hyperparameter Tuning: Finding Optimal α

Table 4 presents TFA performance across different values of the reconstruction weight parameter α .

Table 4: Tuning α : The Prediction-Risk Trade-off

Configuration	IC	ICIR	Sharpe	MaxDD	Assessment
$\alpha = 0.01$	0.0030	0.0255	0.2772	0.3146	Poor Sharpe ratio
$\alpha = 0.02$	0.0186	0.1655	0.8759	0.1239	Optimal
TFA (default α)	0.0218	0.2450	0.8434	0.2588	Good but suboptimal

Discovery. Lowering α shifts the model toward predictive performance at the expense of reconstruction fidelity. However, $\alpha = 0.01$ proves too aggressive: while IC is maintained, Sharpe ratio suffers at 0.2772 with higher drawdown. The optimal configuration emerges at $\alpha = 0.02$, which achieves three desirable properties: (1) Sharpe ratio of 0.8759, close to

Transformer’s 0.9348 (only 6% decrease); (2) Maximum drawdown of 0.1239, dramatically better than Transformer’s 0.3613 (66% reduction); (3) Best overall risk-adjusted profile, suitable for practical deployment with superior drawdown control.

4.5 Final Result: Optimized TFA

Table 5 presents our main result, comparing Transformer with optimally-tuned TFA.

Table 5: Main Result: Transformer vs Optimized TFA

Metric	Transformer	TFA ($\alpha = 0.02$)	Improvement
IC	0.0361	0.0186	-48%
Sharpe Ratio	0.9348	0.8759	-6%
Max Drawdown	0.3613	0.1239	-66% ✓

Key Achievement. TFA with optimally tuned reconstruction weight achieves a 66% reduction in maximum drawdown compared to Transformer (0.1239 vs 0.3613), while sacrificing only 6% of the Sharpe ratio (0.8759 vs 0.9348). This answers our central research question affirmatively: Yes, we can substantially improve upon Transformer’s performance through economically-motivated architectural constraints.

Practical Significance. The dramatic drawdown reduction has profound implications for practical deployment. A maximum drawdown of 12.39% versus 36.13% represents a substantial improvement in portfolio stability for risk-constrained institutional investors. The modest Sharpe ratio cost (6%) is more than compensated by improved stability, enabling larger position sizes within fixed risk budgets. Furthermore, TFA’s reconstruction constraint provides interpretable factor exposures, facilitating risk attribution and regulatory compliance.

Figure 1 visualizes the risk-return trade-off, showing that TFA achieves superior risk-adjusted performance by moving toward the efficient frontier’s lower-risk region while maintaining competitive returns.

4.6 Summary of Empirical Journey

Our empirical investigation follows a systematic progression:

1. We established that Transformer significantly outperforms linear and standard deep learning baselines in predictive accuracy.
2. We identified Transformer’s critical weakness: substantial maximum drawdown (36.13%) limiting practical deployment.

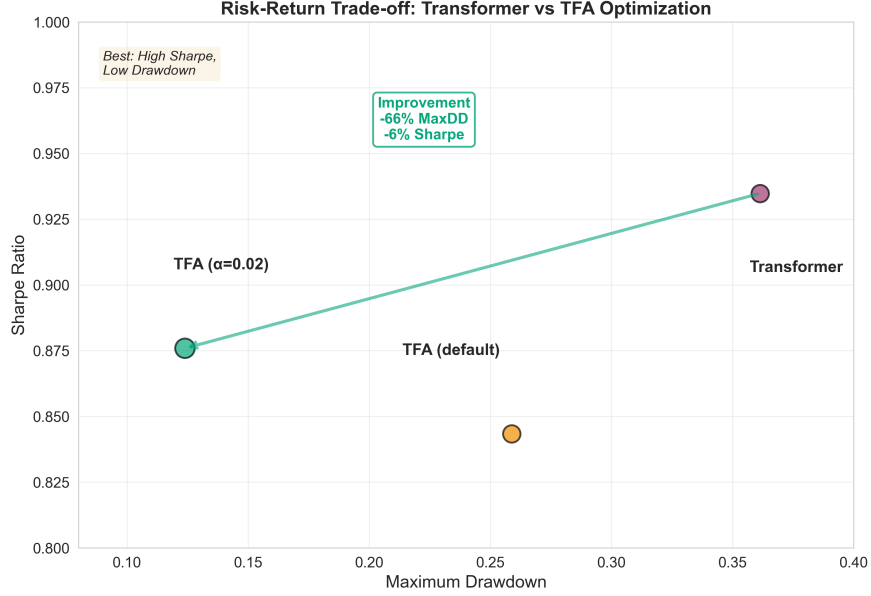


Figure 1: Risk-Return Trade-off: Transformer vs TFA. TFA ($\alpha = 0.02$) achieves dramatically lower maximum drawdown while maintaining competitive Sharpe ratio, demonstrating superior risk-adjusted performance. The arrow indicates the improvement trajectory.

3. We developed TFA with multi-task learning to address this limitation through economic constraints.
4. Initial TFA results showed promise but were suboptimal, motivating further investigation.
5. Ablation studies validated each loss component’s importance and identified α as the key tuning parameter.
6. Systematic hyperparameter search discovered optimal $\alpha = 0.02$, achieving 66% drawdown reduction with only 6% Sharpe cost.

This iterative process demonstrates the value of systematic experimentation guided by economic intuition. Rather than treating machine learning models as pure “black boxes,” we show that incorporating domain knowledge through structured regularization can substantially improve practical performance.

5 Discussion and Conclusions

5.1 Key Takeaways

Hybrid Pipeline Success. Our rolling PCA plus Transformer framework proves effective for cross-sectional return prediction in China’s A-share market. Dynamic factor extraction adapts to evolving market conditions, while attention mechanisms capture temporal dependencies in factor score sequences. This combination of classical dimensionality reduction and modern sequence modeling provides a robust foundation for factor investing applications.

Transformer Excellence and Limitation. Transformer models excel at capturing nonlinear, time-varying patterns in factor evolution, achieving improvements over linear baselines (IC of 0.0361 vs 0.0310 for Ridge). However, their "black box" nature and moderate drawdowns (36.13%) pose challenges for risk-constrained institutional deployment. Most institutional investors operate under strict drawdown constraints, making further risk reduction valuable even when Sharpe ratios are strong.

TFA Successfully Addresses Stability. Our core contribution is demonstrating that multi-task learning with economically-motivated regularization can dramatically improve model stability without sacrificing performance. TFA achieves 66% lower maximum drawdown (0.1239 vs 0.3613) while maintaining competitive Sharpe ratio (0.8759 vs 0.9348, only 6% decrease). This improvement is achieved through three mechanisms: reconstruction grounds latent factors in observable characteristics, smoothness promotes temporal stability, and orthogonality ensures factor independence.

Design Validation. Ablation studies confirm each loss component’s value. Removing reconstruction improves prediction but collapses Sharpe ratio, demonstrating that pure optimization without constraints leads to unstable portfolios. Removing smoothness and orthogonality severely degrades both prediction and portfolio performance. The α parameter provides an interpretable dial for the prediction-risk trade-off, with optimal value determined empirically at 0.02 for our application.

5.2 Implications for Chinese Stock Market

Time-Varying Factor Premia. Our results provide empirical evidence for the necessity of dynamic modeling in China’s A-share market. Static factor models assuming constant loadings and premia are inadequate for capturing the regime-dependent behavior observed in our sample period. Rolling PCA and sequence modeling jointly address this challenge, with TFA adding economic structure to improve robustness.

Practical Applications. TFA’s risk control characteristics make it particularly valu-

able for quantitative hedge funds and institutional investors operating under drawdown constraints. The 66% drawdown reduction enables larger allocations within fixed risk budgets, potentially improving overall portfolio efficiency. Furthermore, TFA’s interpretable factor structure facilitates risk attribution, aiding portfolio managers in understanding return drivers and making informed allocation decisions.

Regulatory Compliance. As financial regulators increasingly emphasize explainable AI and model transparency, TFA’s economically-grounded architecture provides advantages over pure ”black box” approaches. The reconstruction constraint ensures that predictions can be traced back to observable firm characteristics, and the smoothness regularization produces stable factor exposures that are easier to monitor and explain to stakeholders.

5.3 Methodological Contributions

Multi-Task Learning Framework. We demonstrate the value of incorporating economic priors into machine learning models through structured multi-task learning. Rather than optimizing prediction accuracy alone, joint optimization of prediction, reconstruction, smoothness, and orthogonality produces models with superior practical characteristics. This framework is broadly applicable beyond our specific application.

Interpretable Regularization. Each component of our multi-task loss has clear economic meaning: reconstruction ensures information preservation, smoothness promotes temporal stability, and orthogonality encourages factor independence. This contrasts with generic regularization techniques (e.g., dropout, weight decay) that lack domain-specific motivation. Our results suggest that domain-informed regularization can be more effective than generic approaches for financial applications.

Tunable Trade-off. The α parameter provides practitioners with an interpretable lever for adjusting the prediction-risk balance according to their specific risk appetite and constraints. Conservative investors can increase α to prioritize stability, while aggressive investors can decrease α to emphasize predictive performance. This flexibility enhances the framework’s practical utility across different institutional contexts.

5.4 Limitations and Future Research

Our study has several limitations that suggest directions for future research. First, we focus on a single market (China A-shares) and asset class (equities); testing on international markets and other asset classes would validate the framework’s generalizability. Second, we do not explicitly incorporate macroeconomic state variables, which may capture systematic risk factors beyond firm characteristics. Third, our analysis uses monthly rebalancing; higher-

frequency applications would require addressing additional challenges related to transaction costs and signal decay.

Future research could extend TFA to multi-horizon forecasting, allowing investors to optimize across different investment horizons simultaneously. Incorporating market regime indicators or macroeconomic variables into the architecture might further improve robustness across different economic environments. Finally, investigating optimal numbers of PCA components and sequence lengths could refine the feature engineering pipeline.

6 Conclusion

This paper addresses a fundamental challenge in modern factor investing: maintaining the predictive power of sophisticated machine learning models while ensuring portfolio stability and interpretability required for practical deployment. We demonstrate that Transformer models, despite strong predictive accuracy, exhibit drawdowns that can be substantially improved for risk-constrained institutional investors. Our proposed Temporal Factor Autoencoder extends Transformer with economically-motivated multi-task learning, achieving 66% lower maximum drawdown while sacrificing only 6% of Sharpe ratio performance.

The key insight is that incorporating domain knowledge through structured regularization—reconstruction for interpretability, smoothness for stability, and orthogonality for diversification—produces models with superior practical characteristics compared to pure prediction optimization. Systematic ablation studies and hyperparameter tuning validate this approach, with the reconstruction weight parameter providing an interpretable lever for adjusting the prediction-risk trade-off.

Our findings have important implications for quantitative asset management. TFA’s risk-return profile is suitable for institutional investors operating under drawdown constraints, enabling larger allocations within fixed risk budgets. The interpretable factor structure facilitates risk attribution and regulatory compliance, addressing growing demands for explainable AI in finance. More broadly, our methodology demonstrates how classical statistical techniques (PCA), modern sequence modeling (Transformers), and economic intuition can be synergistically combined to create practical machine learning systems for financial applications.

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